

Set B

Question - 1

(a)

$$I = \frac{V_{\max} - V_{\min}}{3R/2 + R + R + R/2} = \frac{16 - 0}{4R}$$

$$\therefore I = \frac{4}{R}$$

$$\therefore V_1 = R/2 \times I = R/2 \times \frac{4}{R} = 2V$$

$$V_2 = (R + R/2) \times I = 3R/2 \times \frac{4}{R} = 6V$$

$$V_3 = (R + R + R/2) \times I = \frac{5R}{2} \times \frac{4}{R} = 10V$$

Quan. range	Quan. level	Encoded binary
0 - 2	1	00
2 - 6	4	01
6 - 10	8	10
10 - 16	13	11

(b) Given, Max^m digital output in the DAC
→ V_{out} at '11' = -13 volts

For R-2R ladder →

$$V_{out} = -\frac{V_{Ref} \times R_F}{2R} \left[b_1 + \frac{b_0}{2} \right]$$

$$\Rightarrow -13 = \frac{-5 \times 26}{2R} \left[1 + \frac{1}{2} \right]$$

$$\Rightarrow R = \frac{-5 \times 26}{-13 \times 2} \times \frac{3}{2}$$

$$\therefore R = 7.5 \text{ k}\Omega$$

∴ Step size of the DAC →

$$V_{out} \text{ (at 01)} = \frac{-5 \times 26}{2 \times 7.5} \left[0 + \frac{1}{2} \right]$$

$$\therefore \Delta_{R-2R} = -4.33 \text{ volts}$$

∴ all output combinations →

binary input ($b_1 b_0$)	analog output
00	0 volts
01	- 4.33 volts
10	- 8.67 volts
11	- 13 volts

(c)

if $V_{in} = 11$ volts given in the

ADC →

it falls under "10-16" Quantization range,

so, 11 volts → 11
in binary

when '11' binary input is given to DAC

so, '11' binary input → -13 volts

(ans)

(d) clock frequency = 4 MHz

$$\therefore \text{Time period, } T_c = \frac{1}{4 \times 10^6} \text{ sec}$$

$$= 2.5 \times 10^{-7} \text{ s}$$

$$= 0.25 \mu\text{s}$$

$$\therefore T_1 = 2^N \times T_c$$

$$= 2^2 \times 0.25 \mu\text{s}$$

$$= 1 \mu\text{s}$$

$$\therefore T_2 = 1.35 \mu\text{s} - 1 \mu\text{s}$$

$$= 0.35 \mu\text{s}$$

Now,

$$T_2 = T_1 \cdot \frac{V_{in}}{V_{Ref}}$$

$$\Rightarrow V_{in} = \frac{T_2 \cdot V_{ref}}{T_1}$$

$$= \frac{0.35 \mu s \times 16 V}{1 \mu s}$$

$$V_{in} = 5.6 \text{ volts}$$

$$\text{Max}^m \text{ conversion time} = 2 \times T_1$$

$$[\text{when, } T_1 = T_2] \leftarrow = 2 \mu s$$

(e) Input voltage range = 16 v

$$\text{number of steps} = 2^N = 2^2$$

$$= 4$$

$$\text{step size} = \frac{16}{4} = 4 \text{ volts}$$

$$V_{in} = m \times \text{step size}$$

$$\Rightarrow m = \frac{V_{in}}{\text{step-size}} = \frac{11}{4} = 2.75 \approx 2$$

'm' value	Q. range	binary
$m = 0$	0 - 4	00
$m = 1$	4 - 8	01
$m = 2$	8 - 12	10
$m = 3$	12 - 16	11

Thus, output of ADC will be '10'
which will be given to R2R DAC →

$$Y_{out}(at\ 10) = \frac{-5 \times 26}{2 \times 7.5} \left[1 + \frac{0}{2} \right]$$

$$= -8.67 \text{ volts}$$

$$\% \text{ error} = \left| \frac{-13 - (-8.67)}{-13} \right| \times 100\%$$

$$\% \text{ error} = 33.3 \%$$

Question-2

(a) The triangular wave generator has
non inverting Schmitt trigger

$$V_{TH} = -\frac{R_1}{R_2} V_L \quad ;$$

$$= -\frac{2}{10} \times (-5)$$

$$= 1V$$

$$V_{TL} = -\frac{R_1}{R_2} V_H$$

$$= -\frac{2}{10} \times (10)$$

$$= -2V$$

$$T_1 = R_i \cdot C_i \left[\frac{V_{TH} - V_{TL}}{V_H} \right]$$

$$= R_i \cdot C_i \left[\frac{1 - (-2)}{10} \right] = \frac{3}{10} R_i \cdot C_i$$

$$T_2 = R_i \cdot C_i \left[\frac{V_{TL} - V_{TH}}{V_L} \right]$$

$$= R_i \cdot C_i \left[\frac{-2 - 1}{-5} \right] = \frac{3}{5} R_i \cdot C_i$$

$$\therefore \text{Time period, } T = T_1 + T_2 = \left(\frac{3}{10} + \frac{3}{5} \right) R_i \cdot C_i$$

$$= \frac{9}{10} R_i \cdot C_i$$

$$\therefore \text{frequency, } f = \frac{1}{T} = \frac{10}{9 R_i \cdot C_i}$$

(Proved)

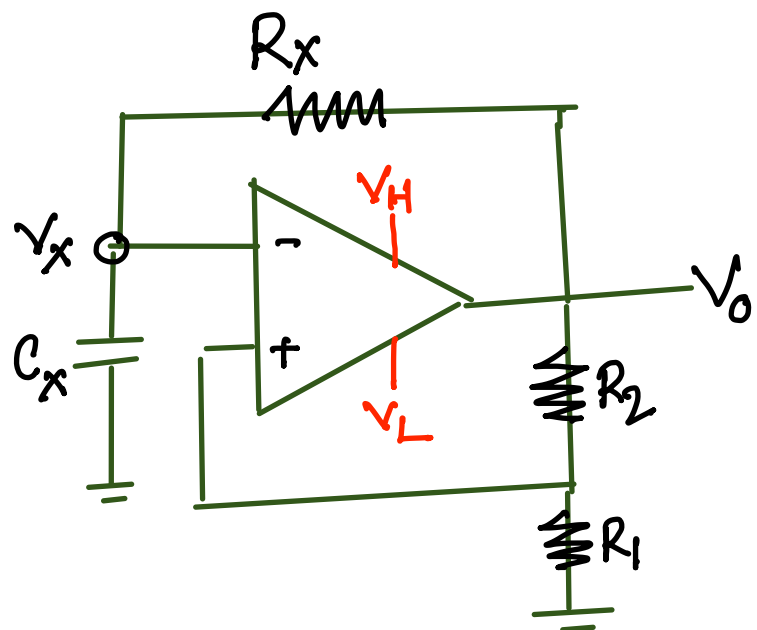
(b) To figure out the duty cycle $\rightarrow$

The triangular wave generator's duty
at $V_B \rightarrow$
 $\downarrow$
is a square
wave

$$\begin{aligned} D.C &= \frac{T_1}{T_1 + T_2} \times 100\% \\ &= \frac{\frac{3}{10} R_1 \cdot C_1}{\frac{9}{10} R_1 \cdot C_1} \times 100\% \\ &= 0.33 \times 100\% \\ &= 33.3\% \end{aligned}$$

This duty cycle will be applied to the
Square wave generator, which is to
be designed $\rightarrow$

$$\begin{aligned} R_1 &= 10k\Omega \\ R_2 &= 20k\Omega \\ R_x &= 10k\Omega \\ C_x &= 50nF \\ f &= 1kHz \end{aligned}$$



$$\therefore \text{Time period, } T = \frac{1}{f} = \frac{1}{1 \times 10^3} \text{ sec} \\ = 1 \text{ ms}$$

$$\therefore \text{High duration, } T_1 = (\text{D.C.}) \times T \\ = 0.33 \times 1 \text{ ms} \\ = 0.33 \text{ ms}$$

$$\therefore \text{Low duration, } T_2 = T - T_1 \\ = 0.67 \text{ ms}$$

$$\therefore \text{Time constant, } T = R_x \cdot C_x \\ = (10 \text{ k}\Omega) \times (50 \text{ nF}) \\ = 5 \times 10^{-4} \text{ sec} \\ = 0.5 \text{ ms}$$

$$\text{Now, } T_1 = T \ln \frac{V_H - V_{TL}}{V_H - V_{TH}}$$

$$\frac{R_1}{R_1 + R_2} = \frac{10}{10 + 20} \\ = \frac{1}{3}$$

$$\Rightarrow \frac{T_1}{J} = \ln \frac{V_H - \left(\frac{R_1}{R_1+R_2}\right) \cdot V_L}{V_H - \left(\frac{R_1}{R_1+R_2}\right) V_H}$$

$$\Rightarrow e^{\frac{0.33}{0.5}} = \frac{V_H - \frac{1}{3}V_L}{V_H - \frac{1}{3}V_H}$$

$$\Rightarrow 1.934 \times \left[\frac{2}{3}V_H \right] = V_H - \frac{1}{3}V_L$$

$$\Rightarrow 1.289 V_H - V_H = -0.33 V_L$$

$$\Rightarrow 0.289 V_H = -0.33 V_L$$

$$\therefore V_L = -0.875 V_H$$

let's assume, positive bias voltage

$$\rightarrow V_H = 10 \text{ V}$$

the negative bias

$$\begin{aligned} \text{voltage, } V_L &= -0.875 \times 10 \\ &= -8.75 \text{ volts} \end{aligned}$$

Question - 3 $\Rightarrow$

(a) From figure-4, we can see that it's an inverting schmitt trigger output,

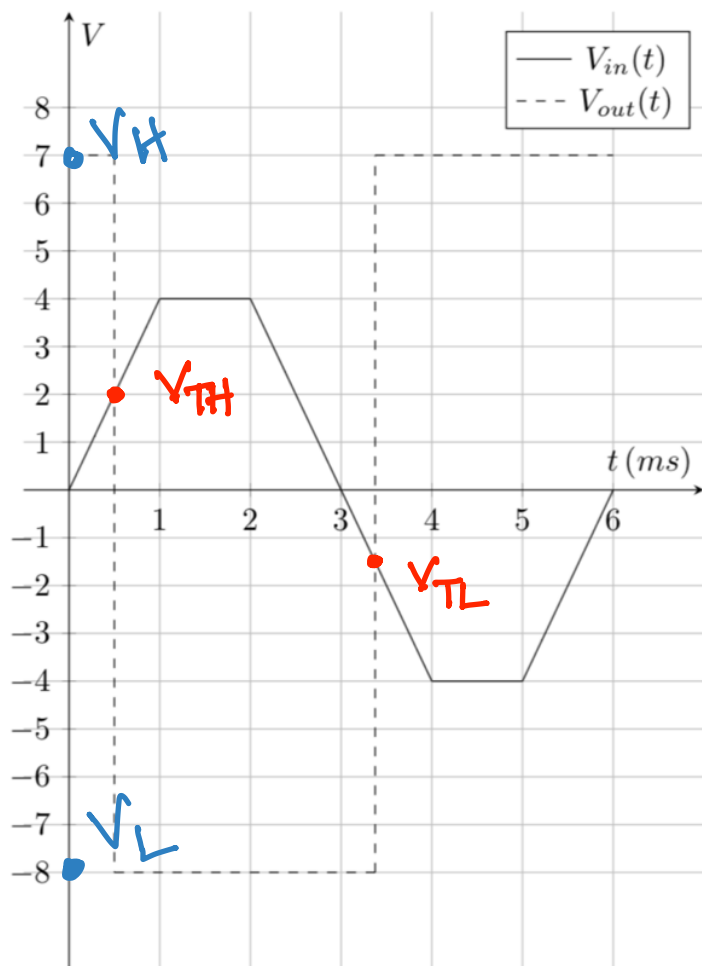


Figure 4: $V_{in}(t)$ & $V_{out}(t)$

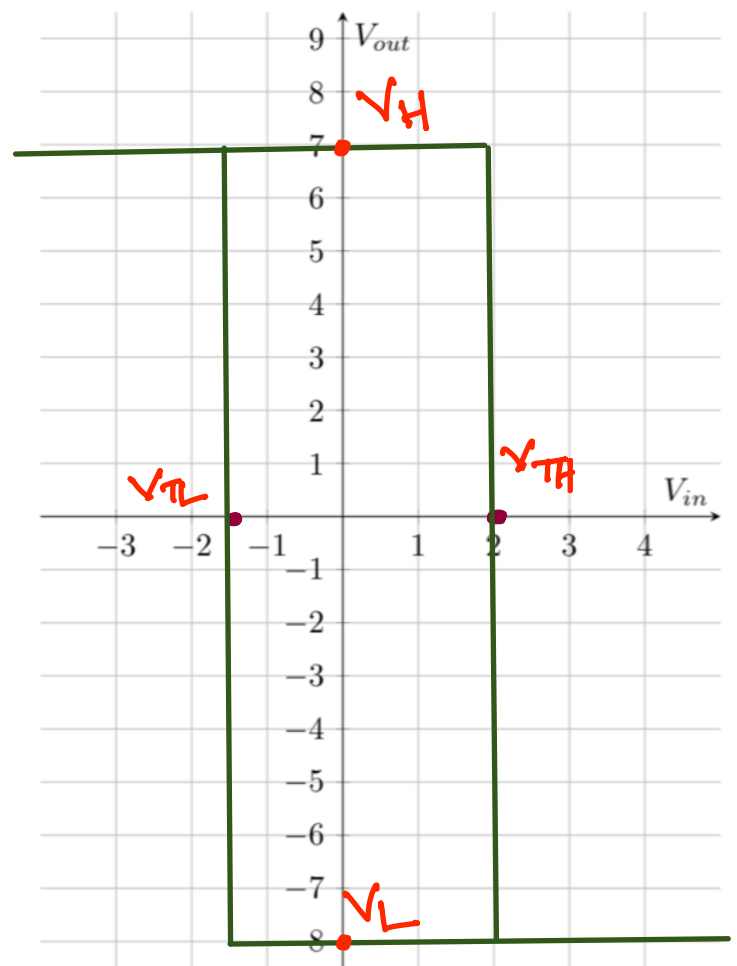


Figure 5: VTC of the Schmitt Trigger

$\therefore$ bias voltages $\begin{cases} V_H = 7V \\ V_L = -8V \end{cases}$

$\therefore$ Threshold voltages $\begin{cases} V_{TH} = 2V \\ V_{TL} = -1.5V \end{cases}$

(b) We know, for inverting Schmitt trigger $\rightarrow$

$$V_{TH} = \frac{R_1}{R_1 + R_2} \cdot V_H + \frac{R_2}{R_1 + R_2} \cdot V_{Ref}$$

shift voltage
' V_s '

$$V_{TL} = \frac{R_1}{R_1 + R_2} \cdot V_L + \frac{R_2}{R_1 + R_2} \cdot V_{Ref}$$

($\rightarrow$) ($-$) ($-$)

$$V_{TH} - V_{TL} = (V_H - V_L) \times \frac{R_1}{R_1 + R_2} + 0$$

$$\Rightarrow \frac{V_{TH} - V_{TL}}{V_H - V_L} = \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow \frac{2 - (-1.5)}{7 - (-8)} = \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow 3.5 R_1 + 3.5 R_2 = 15 R_1$$

$$\therefore R_2 = 3.285 R_1$$

let's assume

$$R_1 = 1k\Omega$$

$$\therefore R_2 = 3.285k\Omega$$

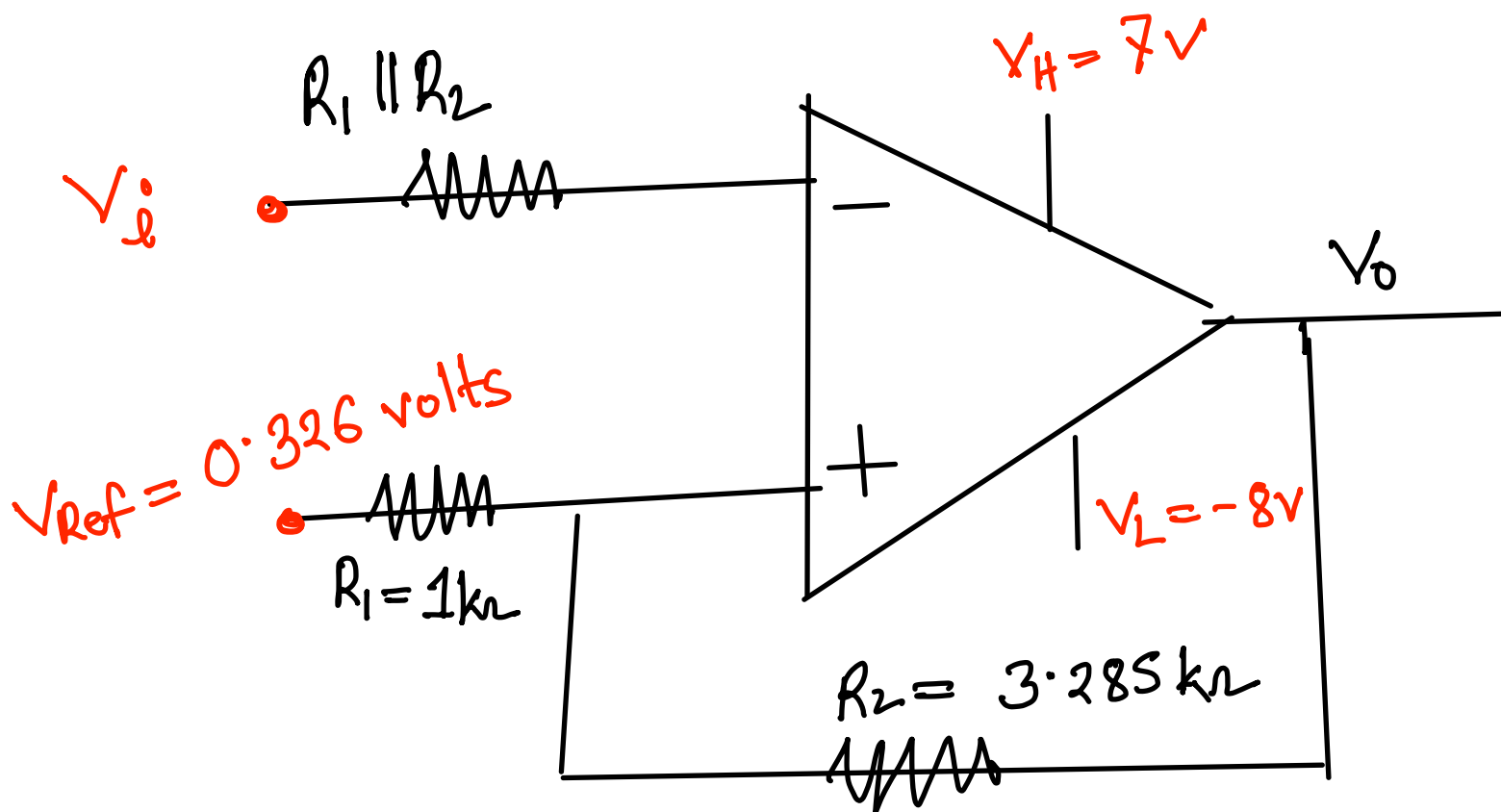
Now,
$$V_S = \frac{R_2}{R_1 + R_2} V_{Ref}$$

$$\Rightarrow \frac{V_{TH} + V_{TL}}{2} = \frac{3.285 R_1}{R_1 + 3.285 R_1} \times V_{Ref}$$

$$\Rightarrow \frac{2 - 1.5}{2} = 0.7667 \times V_{Ref}$$

$$\therefore V_{Ref} = 0.326 \text{ volts}$$

The design $\rightarrow$



(C) Interchanging ' V_i ' and ' V_{ref} ' will give us $\rightarrow$ non-inv. Schmitt trigger

Thus,

$$\begin{aligned} V_{TH} &= -\frac{R_1}{R_2} V_L + \frac{R_1+R_2}{R_2} \cdot V_{ref} \\ &= -\frac{1}{3.285} \times (-8) + \frac{1+3.285}{3.285} \times (0.326) \end{aligned}$$

$$\therefore V_{TH} = 2.86 \text{ volts}$$

$$\begin{aligned} V_{TL} &= -\frac{R_1}{R_2} V_H + \frac{R_1+R_2}{R_2} \cdot V_{ref} \\ &= -\frac{1}{3.285} \times 7 + \frac{1+3.285}{3.285} \times (0.326) \end{aligned}$$

$$\therefore V_{TL} = -0.906 \text{ volts}$$

$$\therefore \text{Hysteresis width, } V_{HW} = V_{TH} - V_{TL} = 3.766 \text{ volts}$$

$$\therefore \text{Shift voltage, } V_S = \frac{V_{TH} + V_{TL}}{2} = 0.977 \text{ volts}$$

Question - 4

(a) Practically, Not Possible

Theoretically,

$$V_{HW} = V_{TH} - V_{TL} \Rightarrow 0 = V_{TH} - V_{TL}$$

$$\therefore V_{TH} = V_{TL}$$

for that, it needs to be

$$V_H = V_L$$

(b) as, $t_1 = T_c \times 2^N$ (all terms are constant)

t_1 doesn't depend on V_{in}

→ that's why remains constant

and, $t_2 = t_1 \cdot \frac{V_{in}}{V_{Ref}}$ → constant

$t_2 \propto V_{in}$ → that's why changes

(c)

Wave 'A' → non-inverting terminal of the Schmitt trigger in triangular wave generator.

Wave 'B' → inverting terminal in square wave generator.